"Harvest Now, Decrypt Later": Threat Modeling, Data Exposure Risk Classification, and Organizational Prioritization Frameworks for the Post-Quantum Era

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***Abstract*—The "Harvest Now, Decrypt Later" (HNDL) threat strategy poses an existential threat to data confidentiality in the transitional period preceding widespread post-quantum cryptography (PQC) deployment. In this threat model, adversaries, particularly nation-state actors, systematically collect encrypted network traffic, cloud data, and communication payloads today, fully intending to decrypt this data once a cryptographically relevant quantum computer (CRQC) becomes available. This paper presents a structured threat model for HNDL threat, categorizes real-world data classes by their exposure risk based on confidentiality lifetime requirements, and proposes a three-tiered organizational prioritization framework for quantum-transition planning. We analyze the asymmetric window between data sensitivity timelines and the anticipated quantum threat horizon (estimated 2030–2035), identify five high-risk data categories (diplomatic communications, financial instruments, healthcare records, intellectual property, and critical infrastructure credentials), and recommend a phased migration roadmap aligned with NIST FIPS 203/204/205 standards and NSA CNSA 2.0 guidelines. Our findings indicate that organizations with long-lived data assets must treat HNDL as an active, present-day threat rather than a future concern.**

***Index Terms*—cryptographic agility, data exposure window, harvest now decrypt later, HNDL threat model, post-quantum cryptography, quantum threat horizon.**

I. Introduction

THE cryptographic infrastructure underpinning modern digital society—public-key encryption, digital signatures, and secure key exchange—rests on mathematical problems believed to be computationally intractable for classical computers. Specifically, the security of RSA and Diffie-Hellman protocols relies on the hardness of integer factorization and the discrete logarithm problem, while Elliptic Curve Cryptography (ECC) depends on the elliptic curve discrete logarithm problem. These assumptions, robust against the most powerful classical hardware, are fundamentally threatened by the advent of quantum computing.

Peter Shor's 1994 quantum algorithm demonstrated, in theory, that a sufficiently powerful quantum computer could solve both integer factorization and discrete logarithm problems in polynomial time [1]. While today's quantum hardware remains far below the threshold required to break production-grade cryptography, the trajectory of quantum engineering advances—particularly in error correction and qubit coherence—suggests that a Cryptographically Relevant Quantum Computer (CRQC) may emerge within the coming decades, with some assessments suggesting a possible horizon in the 2030s [2].

This temporal gap between the present day and Q-Day—the moment a CRQC becomes capable of breaking standard asymmetric cryptography—has given rise to a particularly insidious threat strategy: the "Harvest Now, Decrypt Later" (HNDL) threat. Under this model, sophisticated adversaries intercept and archive encrypted data today, storing it until quantum capabilities mature sufficiently to enable retroactive decryption. While direct public evidence of ongoing HNDL operations is limited, the demonstrated large-scale collection capabilities of nation-state intelligence agencies and the long-term value of encrypted data suggest that such strategies should be considered plausible and strategically significant [8].

The urgency of the HNDL threat is further amplified by the asymmetry between cryptographic migration timelines and data sensitivity windows. Large-scale cryptographic transitions are widely recognized as complex, multi-year undertakings involving technical, operational, and regulatory challenges [4], [10]. Governments, financial institutions, healthcare networks, and critical infrastructure operators generate data with confidentiality requirements spanning 10, 20, or even 50 years. The intersection of these two timelines creates an exposure window that demands immediate action.

This paper makes the following contributions: (1) a formalized threat model for HNDL threat distinguishing it from conventional cyber threats; (2) a taxonomy of five real-world data categories ranked by HNDL exposure risk, incorporating both data lifetime and adversarial capability estimates; (3) a three-tiered organizational prioritization framework for guiding phased PQC migration; and (4) a policy alignment analysis mapping the proposed framework to NIST, FIPS 203/204/205 and NSA CNSA 2.0 mandates.

II. Background and Related Work

A. The Quantum Computing Threat to Cryptography

Quantum computing leverages the principles of superposition and entanglement to process information in fundamentally different ways from classical binary computation. Classical bits exist in one of two states (0 or 1), while quantum bits (qubits) can occupy superpositions of both states simultaneously, enabling massive parallelism in specific computational tasks. Two quantum algorithms are particularly relevant to cryptographic security.

Shor's Algorithm (1994) provides an exponential speedup for integer factorization and discrete logarithm computation. Breaking RSA-2048 is estimated to require roughly 4,000 logical qubits, corresponding to approximately 20 million noisy physical qubits under current error-correction assumptions [5]. Current state-of-the-art quantum hardware, while impressive in raw qubit count, lacks the error correction fidelity required for such operations; however, this is expected to change within a decade.

Grover's Algorithm (1996) provides a quadratic speedup for unstructured search problems, effectively halving the security level of symmetric cryptographic schemes. AES-128, which provides 128 bits of classical security, offers only approximately 64 bits of quantum security under Grover's attack. AES-256, however, retains approximately 128 bits of security—considered sufficient by current standards.

B. The HNDL Threat Paradigm

Awareness of HNDL risks increased as the post-quantum cryptography literature matured [7]. Unlike most cyber threads, HNDL is temporally decoupled: the collection phase (today) and the exploitation phase (future) are separated by years or decades. This separation makes HNDL uniquely difficult to detect and remediate—once data has been harvested, no future cryptographic upgrade can protect it retroactively.

Practical prerequisites for HNDL operations include: (1) passive interception capabilities at network chokepoints (e.g., submarine cables, Internet Exchange Points); (2) large-scale, affordable data storage for archiving petabytes of encrypted traffic; and (3) the eventual availability of a CRQC or quantum decryption service. Nation-state actors with mature intelligence collection infrastructure already satisfy the first two prerequisites [8].

C. NIST Post-Quantum Standardization

In response to the quantum threat, the U.S. National Institute of Standards and Technology (NIST) initiated a Post-Quantum Cryptography standardization competition in 2016. After multiple evaluation rounds, NIST published three final standards in August 2024: FIPS 203 (ML-KEM, based on CRYSTALS-Kyber), FIPS 204 (ML-DSA, based on CRYSTALS-Dilithium), and FIPS 205 (SLH-DSA, based on SPHINCS+) [9]. These standards provide quantum-resistant algorithms for key encapsulation and digital signatures, forming the foundation for post-quantum migration.

The NSA's Commercial National Security Algorithm Suite 2.0 (CNSA 2.0) further mandates timelines for government and defense contractor adoption of PQC algorithms, requiring critical systems to begin migration by 2025 and complete transitions by 2033 [10]. These policy frameworks provide organizational anchors for prioritization decisions.

III. Threat Model

A. Threat Phases and Adversary Capabilities

We model the HNDL threat as a two-phase operation separated by a temporal gap ΔT:

Phase 1 – Collection (Present Day, t_0): The adversary deploys passive interception infrastructure to capture encrypted communications and data at scale. Targets include TLS-encrypted web traffic, VPN tunnels, encrypted email, cloud storage transfers, and database backups transmitted over networks. Encrypted data is stored in compressed archival systems with long retention lifetimes. Collection does not require breaking any encryption; it requires only network access and storage capacity.

Phase 2 – Decryption (Future, $t_0 + \Delta T$): Upon acquiring CRQC access, the adversary retroactively applies Shor's algorithm to recover private keys used to protect the archived data. Decryption is applied preferentially to data with ongoing intelligence value, sorted by classification metadata or traffic analysis inferences made during Phase 1.

To support organizational risk assessment, this paper introduces the Exposure Window (EW) metric, defined as:

$$EW = \max(0, D_m - T_Q) \quad (1)$$

where Dm is the required data confidentiality lifetime (in years from time of creation) and TQ is the estimated number of years until a CRQC becomes operational. When EW > 0, HNDL is a present active threat for that data category.

B. Adversary Profile

HNDL threats require substantial and sustained investment, placing them primarily within the capability set of nation-state actors and well-resourced intelligence agencies. The adversary profile includes: long-term strategic patience; access to fiber-optic tapping infrastructure or legal intercept frameworks; exabyte-scale archival storage; and access to future quantum decryption capabilities (either developed natively or obtained through quantum-as-a-service platforms). Corporate espionage actors with nation-state backing may also participate in collection phases even without immediate decryption capability.

IV. Data Exposure Risk Taxonomy

Not all data is equally at risk from HNDL threats. The exposure risk of a dataset is determined by the intersection of two primary dimensions: (1) the required confidentiality lifetime of the data, and (2) the sensitivity of the information to retroactive disclosure. Based on the literature reviewed and the author's assessment of confidentiality lifetimes and disclosure impacts, this paper proposes five high-risk data categories.

TABLE I
DATA CATEGORY RISK CLASSIFICATION FOR HNDL THREATS

Data Category	Confidentiality Lifetime	HNDL Risk Level	Key Examples
Diplomatic & State Communications	20–50+ years	CRITICAL	Treaties, intelligence cables, military orders
Financial Instruments & Key Material	10–30 years	HIGH	Long-term contracts, payment keys, settlement records
Healthcare & Genomic Records	Lifetime + 50 years	CRITICAL	Patient DNA, psychiatric records, prescription history
Intellectual Property & R&D Data	10–25 years	HIGH	Patent-pending designs, trade secrets, product roadmaps
Critical Infrastructure Credentials	5–15 years	HIGH	SCADA keys, OT network credentials, grid control data
General Enterprise Communications	1–5 years	MODERATE	Internal emails, meeting recordings, project files
Public Web Traffic (non-sensitive)	<1 year	LOW	Public-facing website content, open API responses

A. Category 1: Diplomatic and State Communications

Government communications—diplomatic cables, treaty negotiations, intelligence assessments, and military operational data—represent the highest-value targets for HNDL operations. The confidentiality requirements for such data routinely span decades; some classified documents carry 50-year or permanent classification designations. A CRQC capable of decrypting current-year diplomatic traffic in 2032 would render today's TLS-protected embassy communications fully legible. The geopolitical consequences of such retroactive exposure are potentially catastrophic.

B. Category 2: Financial Instruments and Cryptographic Key Material

Long-term financial instruments, including bond issuances, multi-decade derivative contracts, and settlement ledger records, carry extended confidentiality requirements. More critically, cryptographic key material used to sign financial transactions, if archived during transit or backup operations, could undermine confidence in transaction authenticity, record integrity, and long-term trust in financial systems. The financial sector's heavy reliance on RSA and ECC for PKI makes it acutely vulnerable.

C. Category 3: Healthcare and Genomic Records

Healthcare data presents a unique HNDL risk profile: its confidentiality requirement is effectively unlimited (a patient's genomic data is sensitive for their entire lifetime and potentially that of their descendants). Electronic Health Records (EHR) transmitted over TLS-protected channels today may be archived by adversaries and decrypted years hence, potentially enabling insurance discrimination, coercion, identity misuse, or other forms of long-term privacy harm.

D. Category 4: Intellectual Property and Research Data

Research and development data, proprietary algorithms, patent-pending innovations, and strategic product roadmaps represent high-value targets for state-sponsored industrial espionage. The HNDL model is particularly effective here: an adversary that archives a competitor's or rival nation's R&D communications today can decrypt them within the quantum horizon and exploit the information while the target organization's data is still commercially relevant.

E. Category 5: Critical Infrastructure Credentials

Operational Technology (OT) credentials for power grids, water treatment facilities, transportation networks, and communication backbones are targeted not for their content but for their functional utility. Credential material harvested today and decrypted post-Q-Day could enable destructive cyber-physical threats on infrastructure at a time when defenses may be poorly prepared for quantum-era adversaries.

V. ORGANIZATIONAL PRIORITIZATION FRAMEWORK

Based on the exposure window analysis and data taxonomy, this paper proposes a three-tiered prioritization framework for organizations undertaking PQC migration. The framework is designed to be sector-agnostic while incorporating data sensitivity and operational complexity as primary classification dimensions.

A. Tier 1 – Immediate Action (Now through 2026)

Organizations possessing Category 1 (Diplomatic/State), Category 3 (Healthcare/Genomic), or combinations of high-lifetime data must treat PQC migration as a current operational priority. Recommended actions include:

- **Cryptographic Inventory:** Conduct a full audit of all cryptographic assets, identifying every instance of RSA, ECC, and DH key usage across enterprise systems, APIs, certificates, and embedded devices.
- **Hybrid Encryption Deployment:** Deploy hybrid key encapsulation schemes pairing classical algorithms (e.g., ECDH) with post-quantum alternatives (e.g., ML-KEM per FIPS 203). Hybrid schemes provide quantum resilience without sacrificing classical security during the transition. Experimental evaluations of post-quantum cryptographic integration within TLS environments have demonstrated the practical feasibility of such hybrid deployment approaches [3].
- **Data-at-Rest Classification:** Re-encrypt the highest-sensitivity archived data using AES-256 (quantum-safe under Grover's attack) as an interim measure while PQC key management infrastructure is established [6].
- **Network Traffic Minimization:** Reduce the exposure surface of sensitive data by minimizing plaintext-equivalent metadata in encrypted traffic and adopting forward secrecy for all new TLS deployments.

B. Tier 2 – Strategic Planning (2026–2029)

Organizations with Category 2 (Financial) and Category 4 (IP/R&D) data should use this period to complete cryptographic agility infrastructure, ensuring that future algorithm transitions require minimal operational disruption. Key activities include: completing PKI migration to PQC-compatible certificate hierarchies; piloting ML-DSA digital signatures for document authentication workflows; updating vendor and third-party contractual requirements to mandate PQC capability; and training development and operations teams on quantum-safe library integration.

C. Tier 3 – Compliance and Hardening (2029–2033)

All organizations—including those with Category 5 (Critical Infrastructure) and Category 6 (General Enterprise) data—must complete migration to NIST-standardized PQC algorithms by 2033 in alignment with CNSA 2.0 mandates [10]. This tier encompasses full deprecation of RSA and ECC across all enterprise systems, completion of code signing and software supply chain PQC integration, and independent cryptographic audit and certification.

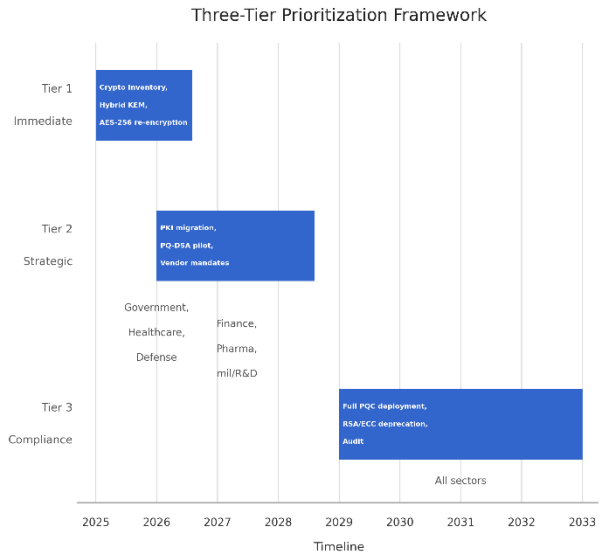


Fig. 1. Three-tier prioritization framework summary.

VI. Policy Alignment Analysis

The proposed framework aligns substantively with existing government policy directives. NIST SP 800-208 [11] and the subsequent FIPS 203/204/205 publications [9] provide the algorithmic foundation for migration. NSA CNSA 2.0 [10] provides the compliance timeline. Additionally, the White House National Security Memorandum NSM-10 (2022) [12] explicitly identified HNDL as an active threat and mandated federal agencies to begin quantum-resilience planning immediately.

The European Union Agency for Cybersecurity (ENISA) has published complementary guidance emphasizing risk-based approaches to PQC migration [13], broadly consistent with our tiered framework. Financial regulators are increasingly emphasizing operational resilience and long-term cyber-risk management, principles that are relevant to future quantum-related security challenges [14].

A key policy gap identified in our analysis is the absence of mandatory HNDL disclosure requirements: organizations that have likely been subjected to large-scale encrypted data collection have no legal obligation to disclose this risk to data subjects. We recommend that data protection regulators consider HNDL as a qualifying risk event under existing breach notification frameworks, requiring organizations that hold long-lived sensitive data to disclose their quantum readiness posture.

VII. Discussion

A. Limitations and Uncertainty

The primary uncertainty in HNDL threat modeling is the Q-Day timeline. Estimates regarding the arrival of a cryptographically relevant quantum computer vary considerably, with some assessments suggesting emergence within the next decade and others projecting a substantially longer timeframe [2], [15]. This uncertainty does not diminish the urgency for high-lifetime data categories (Tiers 1 and 2); even under the most conservative Q-Day estimate, data generated today with a 25-year confidentiality requirement falls within the CRQC horizon. For lower-lifetime data (Tier 3), the uncertainty is more consequential and warrants scenario planning across multiple Q-Day assumptions.

B. The Role of Cryptographic Agility

A central architectural recommendation arising from HNDL threat modeling is the concept of cryptographic agility: designing systems such that cryptographic algorithms can be swapped without systemic redesign. Organizations that hard-code RSA key sizes or specific cipher suites into embedded devices or long-lived software systems will face prohibitive migration costs. Cryptographic agility, while often deprioritized under cost pressure, is the single most impactful architectural investment an organization can make today against the HNDL threat.

C. Quantum Key Distribution vs. Post-Quantum Cryptography

Quantum Key Distribution (QKD) is sometimes proposed as an alternative to PQC for addressing the quantum threat. While QKD provides information-theoretic security for key exchange, it is impractical for most organizational deployments due to its dependence on dedicated fiber-optic infrastructure, limited range, and inability to be deployed over existing IP networks. PQC, by contrast, is a software-based solution compatible with existing network infrastructure. For organizational transition purposes, PQC is the operationally viable path; QKD may complement PQC in high-security point-to-point contexts.

VIII. Conclusion

The "Harvest Now, Decrypt Later" (HNDL) threat represents a present-day security challenge driven by the gap between long-term data confidentiality requirements and the future capabilities of quantum computing. This paper formalized the HNDL threat model, proposed a five-category data exposure taxonomy, and introduced a three-tiered framework to support organizational post-quantum cryptography (PQC) migration. A key finding is that cryptographic risk assessment should focus not on when quantum computers will arrive, but on whether data generated today must remain confidential beyond Q-Day. With PQC standards and migration guidance now established, organizations should begin building quantum resilience through cryptographic inventory, hybrid encryption deployment, and agility-focused system design to preserve long-term data confidentiality and trust in the post-quantum era.

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